A Demand-Driven Framework for India's AI Infrastructure Planning: Linking Compute, Energy, Physical Resources, and Semiconductor Dependencies

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Abstract

National AI strategies increasingly treat high-performance compute accelerators as primary instruments of technological sovereignty. However, compute scaling is physically bounded by electricity transmission, cooling water, semiconductor lithography, advanced packaging, and geopolitically concentrated supply chains. This paper develops an evidence-backed **Five-Pillar Conceptual Framework** integrated with an auditable six-layer **Demand-Cascade Model** to evaluate India's AI infrastructure trajectory across 200k, 500k, and 1M GPU deployment scenarios. For the baseline 500k GPU case, the model projects peak facility power c... India's operational colocation capa... 6.25 TWh (4.475 Mt CO2), and wat... ML/year. Crucially, empirical results ... hypothesis that front-end fab ca... chokepoint: wafer starts represent ... sub-5nm output, whereas CoWoS packaging absorbs 7.41% (4.6x higher intensity) and HBM3e consumes 6.67% of global supply. Policy planning must invert standard priorities.

Keywords: AI infrastructure, semiconductor sovereignty, demand-cascade modeling, power grid constraints, CoWoS packaging, HBM.

I. INTRODUCTION

AI has transitioned from an algorithmic discipline to an ultra-compute-intensive industrial apparatus governed by capital expenditure, high-voltage power grids, and physical microelectronic supply lines. India's IndiaAI Mission (Rs. 10,371.92 crore, ~$1.25B) and Semicon India 2.0 (Rs. 1,27,500 crore) codify this ambition.

Prevailing policy discourse decouples accelerator procurement from the physical systems required to sustain compute operations. A 500k-GPU cluster demands continuous megawatt-scale power, high-voltage substations, gigalitres of cooling water under tropical ambient temperatures, and supply chains spanning EUV lithography, CoWoS packaging, and HBM. Without rigorous co-planning, fiscal subsidies risk procuring silicon that sits idle behind multi-year transmission queues or creates municipal water conflicts.

Middle-power economies face the *N-2 Problem*: non-superpower nations cannot realistically build commercial sub-3nm foundries (>$20B capex/fab) without crippling under-utilization. This paper introduces an integrated Five-Pillar Framework and an auditable six-layer Demand-Cascade Model to identify true structural chokepoints.

II. LITERATURE REVIEW AND RESEARCH GAP

A. Data Centre Thermodynamics and Grid Scaling

Shehabi et al. [LBNL 2024] document server-level host overheads multiplying GPU TDP by 1.6-2.1x. IEA [2024] projects global data centre electricity demand exceeding 1,000 TWh by 2026. JLL and CBRE confirm India's operational colocation capacity at 1,575 MW (mid-2026), with 52.4% concentrated in Mumbai and substation interconnection queues of 18-36 months.

B. Semiconductor Geopolitics and Foundry Chokepoints

Akimitsu [2026] demonstrates extreme market concentration (HHI: EUV 10,000; refined gallium 9,804; CoWoS packaging 8,100). SemiAnalysis and TrendForce identify CoWoS 2.5D packaging as the primary AI hardware bottleneck, with HBM3e concentrated in a tight triopoly (SK Hynix, Samsung, Micron).

TABLE I. Prior Art Benchmark and Analytical Comparison Matrix

Source	Methodology	Scenarios	Phys. Infra	Micro-HW
Akimitsu 2026	Econometric HHI	None	None	HHI 40 stages
Portulans 2026	Qualitative policy	Conceptual	None	Taxonomy
IEA / LBNL 2024	Macro energy	Global TWh	PUE 1.15-1.25	None
Inherited Draft	Heuristic Vuln.	10k/45k	Qual. PUE 1.55	Cites HHI
This Paper v2.0	**Demand Cascade**	**200k/500k/1M**	**MW,TWh,Water**	**CoWoS,HBM**

C. Five Core Research Gaps

Five critical research deficits remain unaddressed:

(1) **RG-01 Disciplinary Siloing:** Microelectronics, energy, and governance research are disconnected.

(2) **RG-02 Absent Demand Cascade:** No study propagates GPU targets (200k-1M) bottom-up through chip TDP, PUE, wafer geometry, and packaging.

(3) **RG-03 Silicon Wafer Fallacy:** Policy assumes domestic sub-5nm fab is the binding constraint without testing whether packaging limits deployment first.

(4) **RG-04 Tropical Climate Blind Spot:** Global models assume temperate PUE baselines (1.15-1.20), ignoring Indian summer heatwave spikes (>42C, PUE up to 1.65).

(5) **RG-05 Policy Operationalization Gap:** No quantitative intervention matrix distinguishing build vs. partner vs. manage.

III. DEMAND-CASCADE MODEL FORMULATION

The model maps an exogenous accelerator target N through six deterministic physical layers, fully parameterized from empirical data centre audits, foundry specifications, and CEA telemetry.

A. Layer 1: IT Electrical Load

GPU TDP P_gpu = 700 W (H100 SXM5); chassis overhead R_host = 1.82 (CPUs, DDR5, InfiniBand, PCIe):

$$P_{IT} = N \times P_{gpu} \times R_{host} / 10^6 \ [MW]$$

B. Facility Power and Annual Energy

P_facility = P_IT x PUE (tropical baseline PUE = 1.40)

E_annual = P_facility x 8760 x 0.80 / 10^6 [TWh] (80% utilization)

Emissions: E_annual x 0.716 tCO2/MWh [Mt CO2] (CEA grid factor)

C. Layers 3-6: Water, Racks, Wal

Water: I_water x E_annual x 10^9 / 10^3 / 60 kW.

Wafer starts: Gross dies = 63 (GH1 wafer, edge-loss eq.). Murphy (D0=0.08/cm2); industrial net yie dies/wafer. W_starts = ceil(N/40).

CoWoS interposers: W_CoWoS = N_HBM = N x 8.

TABLE II. Demand-Cascade Sir Scenarios (Mod

IT MW	Fac MW	TWh/yr	Water ML					
224	280	1.59	319					
255	357	2.50	4,500					
420	651	5.13	12,832					
560	700	3.99	797					
637	892	6.25	11,250					
1,050	**1,628**	**12.83**	**32,080**					
1,120	1,400	7.97	1,594					
1,274	1,784	12.50	22,499	25,000	3.21%	66,667	33.74%	200.67%
2,100	3,255	25.66	64,160	52,632	6.75%	83,334	18.52%	200.67%

Note: 500k Base row highlighted. CoWoS% of global ~450k wafers/year; TSMC% of ~780k sub-5nm wafers/year; Colo% of India 1,575 MW operational capacity.

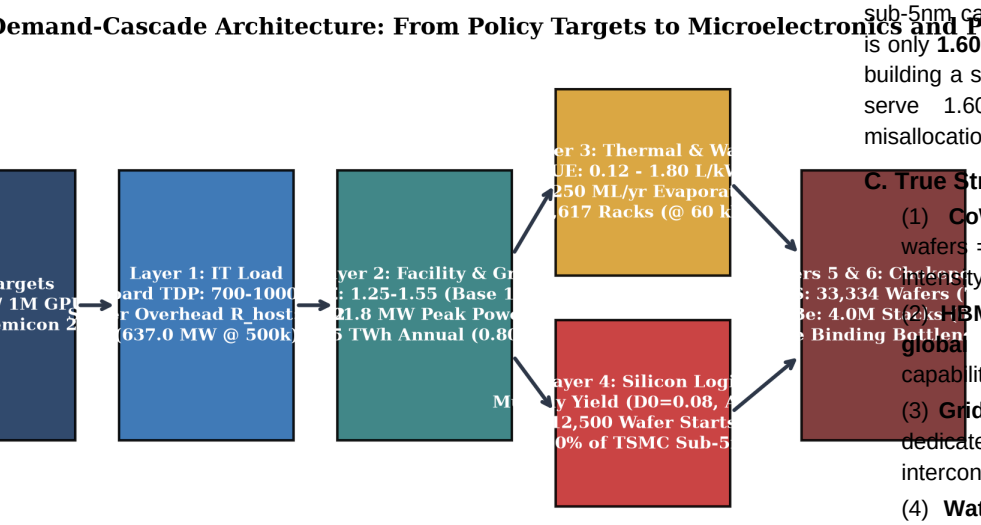


Fig. 1. *Six-Layer Demand-Cascade Architecture: From Policy Targets to Microelectronics and Power Grids.*

A. Scenario Scaling Analysis

In the baseline 500k-GPU scenario, IT load reaches 637.0 MW. Under tropical PUE = 1.40, peak facility power escalates to **891.8 MW — 56.62% of India's entire operational colocation capacity**. Annual electricity totals 6.25 TWh/year (4.475 Mt CO2); evaporative water dissipation reaches 11,249.5 ML/year (11.25 billion litres).

Fig. 2. The Structural Chokepoint Inversion (500,000-GPU Baselin

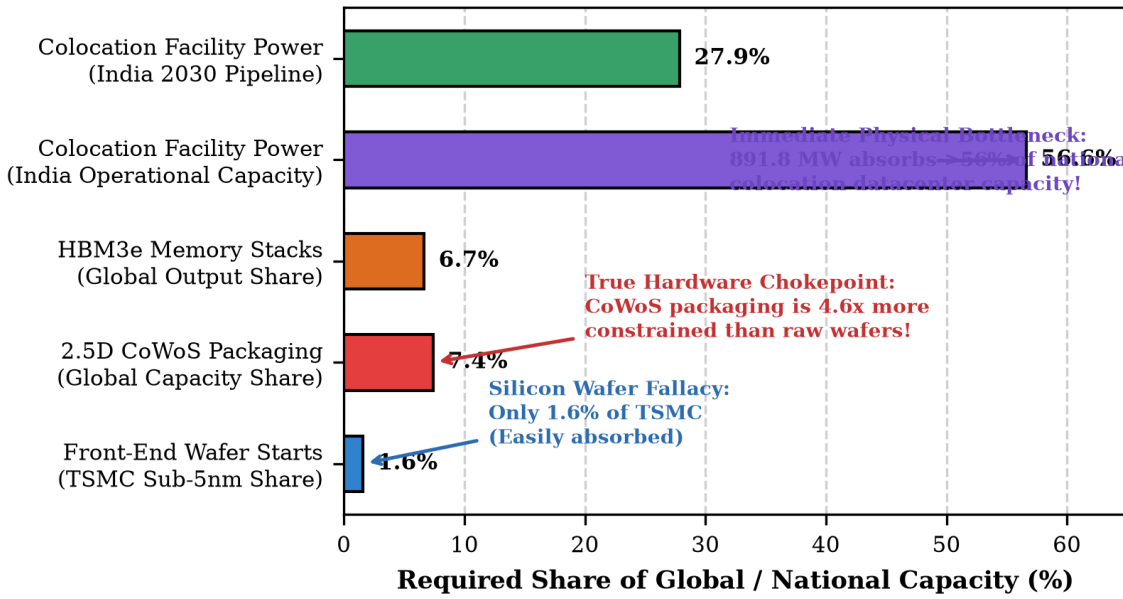


Fig. 2. The Structural Chokepoint Inversion: CoWoS packaging (7.41%) and grid capacity (56.62%) are binding constraints — not front-end wafers (1.60%). Red bars denote binding chokepoints.

B. Refuting the Silicon Wafer Hypothesis

Deploying 500k GPUs requires 12,500 wafer starts. TSMC sub-5nm capacity is ~780,000 wafers/year. India's requirement is only **1.60% of annual global sub-5nm output**. Domestically building a sub-5nm gigafab ($15-20B capex, 7-10 year lead) to serve 1.60% of demand constitutes a severe capital misallocation.

C. True Structural Chokepoints

(1) **CoWoS Advanced Packaging:** 33,334 interposer wafers = **7.41% of global capacity** (4.6x higher bottleneck intensity than front-end wafers).

(2) **HBM3e Memory:** 4,000,000 HBM stacks = **6.67% of global supply**. India has zero domestic fabrication capability.

(3) **Grid Congestion:** 891.8 MW into Mumbai requires 18 dedicated 50 MW substations, triggering 24-36 month interconnection delays (CEA guidelines).

(4) **Water Stress:** 11.25 billion litres/year conflicts with urban water supply.

TABLE III. Semiconductor vs. Physical Infrastructure Chokepoint Hierarchy

Infrastructure Layer	HHI	India	500k Sha
EUV Lithography (ASML)	10,000	0%	Upstrear

IV. El

	HHI	
	5,400	
	8,100	
	4,950	
	2,200	
	N/A	

s-National AI Infrastructure & Str

ted Accelerated Compute Fleet

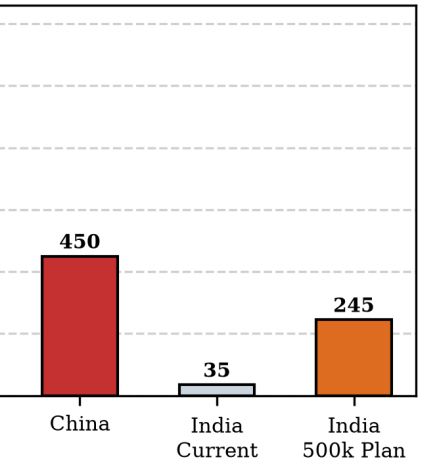




Fig. 3. *Cross-National AI Infrastructu stocks and asymmetric supply-chain China, and In*

D. Cross-National Benchmarking

India hosts ~20% of global chip d GCCs but possesses 0% domestic packaging, and 0% HBM fabricatic elevates India's active compute fr approaching China's 450 EFLOPS, b foreign dependencies across all supp

E. Cooling Technology Pareto Tr

In summer heatwaves (>42C), traditi PUE = 1.65 and water consumption Direct-to-Chip (D2C) liquid cooling r by **94.6%** (11,651 to 631 ML) and MW.

Fig. 4. Cooling Technology Water-Energy Pareto Frontier under Tropical Indian A

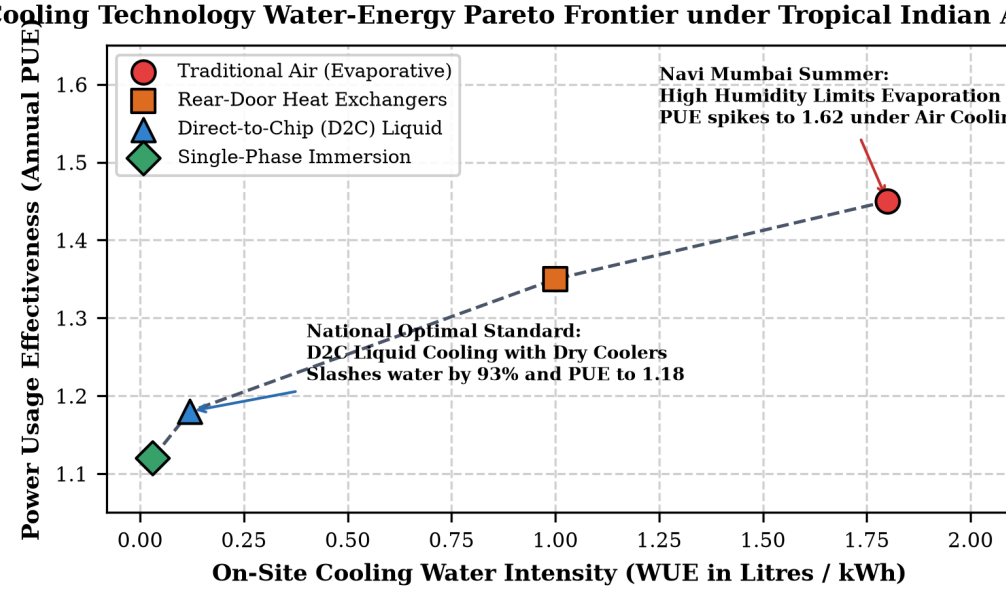


Fig. 4. *Cooling Technology Water-Energy Pareto Frontier across Indian Climatic Profiles. D2C liquid cooling dominates on both PUE and water metrics.*

Fig. 5. One-at-a-Time Parameter Sensitivity & Uncertainty Torna

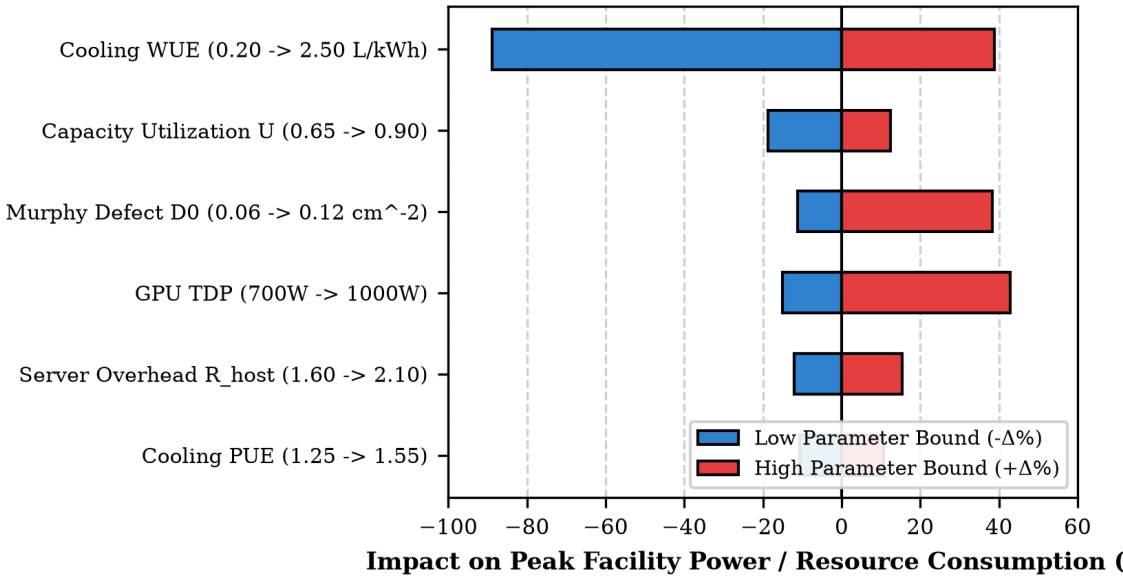


Fig. 5. *One-at-a-Time Parameter Sensitivity (Tornado) and Monte Carlo Uncertainty Analysis (1,000 runs). 90% CI for facility power: [834 MW, 1,130 MW].*

TABLE IV. Cooling Technology Trade-Off Matrix — Tropical Indian Ambient (500k GPUs)

	Ann. PUE	Sum. PUE	WUE L/kWh	Water ML
	1.45	1.65	1.80	11,651
	1.35	1.45	1.00	6,032
	1.18	**1.25**	**0.12**	**631**
	1.12	1.18	0.03	151

F. Sensitivity and Monte Carlo Analysis

Tornado analysis (Fig. 5) confirms facility power is most sensitive to PUE and GPU TDP. Switching from air cooling (PUE=1.55) to D2C (PUE=1.18) saves **235.7 MW** of peak grid demand at 500k scale. Monte Carlo 90% CI: facility power [834-1,130 MW], wafer starts [11,112-14,706].

V. STRATEGIC ROADMAP AND POLICY IMPLICATIONS

A. Priority Inversion: Debunking Foundry-First

Standard policy advocates $15-20B for a domestic sub-5nm foundry. The cascade model proves this is a strategic misallocation: 12,500 wafers/year (< 2% of a gigafab's break-even volume) produce zero operational accelerators without CoWoS, HBM, and 891.8 MW of substation power.

Old priority: Sub-5nm logic fab ($15B+ capex, 7-10yr lead). **New priority:** High-voltage Green AI transmission corridors, spatial decentralization, and bilateral packaging/memory JVs.

B. Three-Tier Strategic Intervention Matrix

Tier 1 — Build Domestically (2026-2028):

- Mandate 15-20% capital rebates directing greenfield AI clusters to Gujarat, Rajasthan, Odisha, MP — reducing Mumbai concentration below 35%.
- Condition IndiaAI subsidies on PUE <= 1.25 and closed-loop dry heat rejection, eliminating 11 billion litres/year of municipal water loss.
- Fast-track 9-month CEA approval window for 400/220 kV substations serving clusters >= 50 MW.

Tier 2 — Partner Strategically (2028-2031):

- Expand ATMP/OSAT facilities (Tata Electronics, Micron) toward 2.5D silicon/glass interposers via JVs with Taiwanese/Japanese OSATs under Semicon 2.0.
- Scale CSIR-IMMT gallium extraction from NALCO/Hindalco bauxite residue to 50 MT/year.
- Mobilize 120,000 VLSI engineers into an open-standard (UCIe) chiplet consortium for domain-specific RISC-V matrix accelerators on mature nodes.

Tier 3 — Manage External Dependencies (2031-2035):

- Secure multi-year sovereign foundry agreements at TSMC and bilateral HBM supply treaties with South Korea, leveraging India's consumer market as strategic trade offsets.

VI. CONCLUSION AND LIMITATIONS

This paper presents the first reproducible, multi-layer demand-cascade framework linking national AI accelerator targets to electric transmission, cooling thermodynamics, and semiconductor supply chains. The empirical findings prove that India's sovereign compute bottleneck is **2.5D CoWoS packaging**, **HBM allocation**, and **metropolitan grid congestion** — not front-end lithography. By prioritizing transmission corridors, spatial dispersion, and packaging alliances, India can deploy resilient sovereign AI infrastructure without capital misallocation.

Limitations: Model assumes static 80% utilization without dynamic nodal power dispatch, and relies on public foundry defect density disclosures. Future work will integrate real-time AC power-flow simulations and dynamic diurnal inference scaling.

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